Ruihong Shen 沈睿弘

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https://ruihong04.github.io

Education

School of EECS, Peking University — Beijing, China
B.Sc. in Artificial Intelligence — 2023 - 2027 *(expected)*

- Admitted to Zhi Class, Honor class lead by Prof. Baoquan Chen
- GPA: 3.87/4.0; Rank: 1/70

Research

* Joint first author † Corresponding author

CCVL Lab | Department of CS, Johns Hopkins University — 2026.06 - 2026.08
Visiting Student; advised by Prof. Alan Yuille and Dr. Jieneng Chen

- **4DCodeBench: Benchmarking Agents on Inverse Graphics of Dynamic Scenes**
  Ruihong Shen*, Žiga Kovačič*, Peter Kulits*, Xingrui Wang, Zizhang Li, Joshua B. Tenenbaum, Alan Yuille, Jieneng Chen‡, Jiajun Wu‡
  arXiv 2026 *Tags: Inverse graphics, VLM, physics simulation, Coding world model*

CoRe Lab | Institute for AI, Peking University — 2024.04 - Present
Advised by Prof. Yixin Zhu

- **Neural Force Field: Few-shot learning of generalized physical reasoning**
  Shiqian Li*, **Ruihong Shen***, Yaoyu Tao†, Chi Zhang†, Yixin Zhu†
  ICLR 2026 *Tags: Intuitive physics, representation learning, abstract physical reasoning*

- **Learning Physics-Grounded 4D Dynamics with Neural Gaussian Force Fields**
  Shiqian Li*, **Ruihong Shen***, Junfeng Ni, Chang Pan, Chi Zhang†, Yixin Zhu†
  ICLR 2026 *Tags: Intuitive physics, dynamics prediction, world models*

- **Overhang Tower: Resource-Rational Adaptation in Sequential Physical Planning**
  Ruihong Shen, Shiqian Li, Yixin Zhu†
  CogSci 2026 *Tags: Intuitive physics, Cognitive Science, Psychology*

- **A sub-10ms neural dynamical system based on phase change memristors**
  Lei Cai*, Yaoyu Tao*†, Chenchen Xie*, Longhao Yan*, Shiqian Li, **Ruihong Shen**, Zelun Pan, Xile Wang, Bowen Wang, Daijing Shi, Yihang Zhu, Teng Zhang, Yixin Zhu†, Xi Li, Zhitang Song†, Ru Huang, Yuchao Yang†
  Science 2026 *Tags: Neural ODE, Hardware-software co-design*

Selected Awards

- **Sensetime Scholarship** (selects 30 Chinese AI undergrads annually), 2026.06
- **CogSci Student Travel Grant** (45 person / year), 2026.05
- **National Scholarship** (highest honor for Chinese undergrads), 2024.12

Services

Teaching Assistant: Data Structure and Algorithm (STEM students), Spring 2025

Skills

Languages: Mandarin (native), English (TOEFL iBT: 6.0/6.0; S: 5.5).
Programming: `Python, C/C++, CUDA`
Tools: `PyTorch, Blender, Mujoco`

Research Interests

`4D`, `Physical reasoning`, `World models`